



Introduction

What is a green roof?

A green roof, or rooftop garden, is a vegetative layer grown on a rooftop.

What are the benefits of a green roof?

They provide shade, remove heat from the air, and reduce temperatures of the roof surface and surrounding air. Using green roofs in built environments with limited vegetation can moderate the heat island effect, particularly during the day.

Which buildings on campus already have a green roof?

UMBC has several green roofs. Patapsco Hall, The Interdisciplinary Life Sciences Building (ILSB), Administration Building, Chesapeake Employers Insurance Arena, and Apartment Community Center.

Our question:

Which UMBC buildings have low-slope, relatively uncluttered, sun-exposed roof areas large enough to make green roofs practical and beneficial?

Existing green roofs

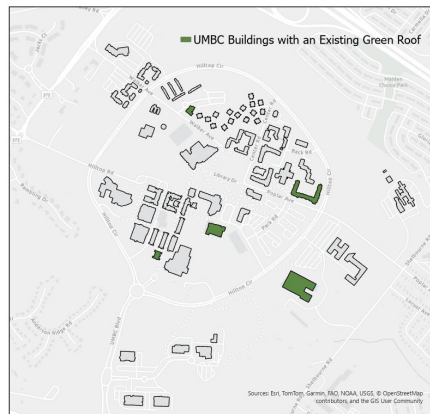


Fig. 1. Existing green roofs have flatness percentage of ~54%

Methodology

Data: UMBC LiDAR and UMBC aerial photography (2024) used as visual/reference layers.

Surfaces: LiDAR processed to DEM (ground) and DSM (top of roofs); nDSM = DSM – DEM for building heights.

Analyses: Computed slope, estimated solar radiation/shade (June 1st-September 30th, 2025), and mapped clutter/obstructions (HVAC, vents, skylights), emphasizing contiguous open roof area.

Suitability & evaluation: Standardized clutter and solar radiation layers. Combined to a suitability index, and summarized by building for ranking and map display.

References

Moniz de Sa, Sarah, 2025, "Assessment of Green Roof Suitability by Active Remote Sensing of University of British Columbia Buildings", <https://doi.org/10.5683/SP3/LVXF0V>, Borealis, V1

Surface Model Generation and Slope Analysis

Used raw LiDAR data to create a DEM and DSM. Then, produced a normalized DSM by subtracting the DEM from the DSM. Slope tool used to quantify output in degrees.

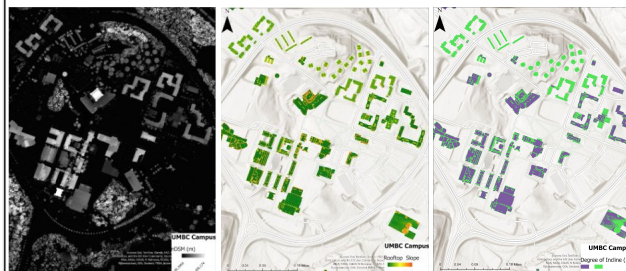


Fig. 2. Pixels with slopes less than or equal to the threshold range ($\leq 15^\circ$) were coded as 1 (suitable) and greater than were coded as 2 (unsuitable). This created a binary slope-suitability raster.

Solar Radiation

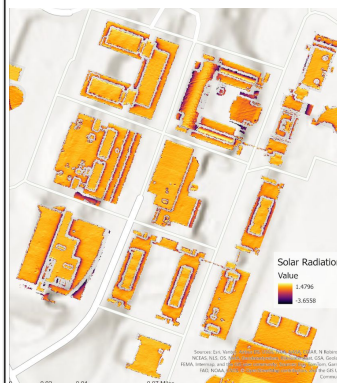


Fig. 3. Solar Radiation

- ArcGIS Raster Solar Radiation Tool: Solar Radiation Energy (kWh/m^2) for June - Sept (growing season). DSM layer used as input raster.
- Outlier values e.g. at walls or obstacles were masked out via a determined threshold value
- Standardized raster via z-score normalization: $\text{pixel} = (X - \mu) / \sigma$... (value - mean)/SD
- Standardizing allows comparability of different types of data and data ranges

Standard deviation (variability) of roof elevation values was used as an estimate of clutter. Roofing with significant obstacles e.g. HVAC units were expected to have > variable elevation values.

- DSM $\rightarrow$ Focal statistics (4x4) of standard deviation
- Outlier values were masked out via a determined threshold value
- Standardized raster via an inverse standardization: -1 (previous formula). Makes high clutter values subtract rather than add to suitability

Clutter

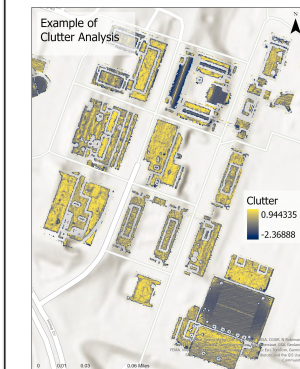


Fig. 4. Standardized clutter and solar radiation layers were added. Building shapefile was modified to include only those between 30% and 80% flatness. This shapefile was used to mask the combined raster, creating the suitability index layer. Zonal histogram $\rightarrow$ median used to create building median suitability scores and corresponding bar chart.

Results

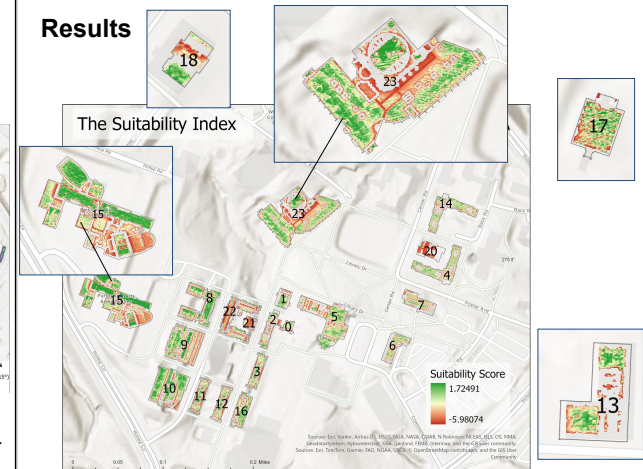


Fig. 5. Notable roof levels (not considered in building median): 15, 10, 8

Overview of final included building locations

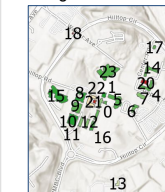


Fig. 6. Building Median Suitability

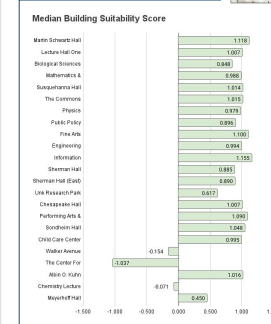
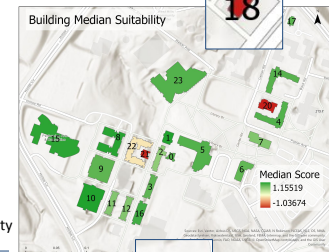


Fig. 7 Final Suitability Score

Conclusions

Performing Arts and Humanities Building (15), Info. Technology/Engineering Building (10), and Fine Arts Building (8) have the highest suitability for green roofs. They have large areas with no clutter and low slope but enough for drainage. Martin Schwartz Hall (1) is seen to have the highest score in Figure 7 but is quite a small building compared to the other three and is lacking the area needed.



ID #	Building
0	Martin Schwartz Hall
1	Lecture Hall One
2	Biological Sciences
3	Mathematics & Psychology
4	Susquehanna Hall
5	The Commons
6	Physics
7	Public Policy
8	Fine Arts
9	Engineering
10	Information Technology/Engineering
11	Sherman Hall
12	Sherman Hall (East)
13	Unit Research Park Building
14	Chesapeake Hall
15	Performing Arts & Humanities
16	Southern Hall
17	Child Care Center
20	The Center For Web-Being
22	Meyershoff Hall
23	Allen D. Kulin Library and Gallery

Fig. 8 Building ID Legend